

Engagement Coding Manual

Novel Word Learning from Social Robots by Autistic and Non-Autistic Children

This document summarizes the scheme used to code child engagement from video. The authoritative version is the full coding guide (Engagement Coding Template.docx) and its accompanying coding spreadsheet, both deposited in this folder. The scheme was adapted from the engagement paradigm used in the authors' prior work. This summary is provided for quick reference and is not a substitute for the full guide and its worked examples.

Unit of analysis

Codes are entered for each 10-second block of the robot interaction. Blocks are numbered chronologically from 1 (for example, block 12 covers seconds 111-120). Coding spans three task segments: introduction and playtime, question and answer, and the after-task / goodbye segment, with the start and end of each segment anchored to specific points in the robot's script.

Missing-data value (99)

A value of 99 denotes a behavior that was precluded from being performed or observed, rather than a low score. Examples include a child who was asked not to touch the robot (coded 99 on the tactile channel from that point onward) and a child who is not visible on the video but has not left the room (coded 99 for the affected blocks). The 99 value is treated as missing when aggregate engagement scores are computed. (For the emotional channel, 99 is used only when the child is not visible in frame.)

Behavioral engagement channels

Behavioral engagement is coded on four channels, each on a 1 to 5 scale where 5 is the highest engagement and 1 indicates absent, non-compliant, negative, or evasive engagement.

Proxemic engagement (where the child is in space relative to the robot)

Level	Meaning	Description
5	High	Touching the robot or right next to it (within ~10 cm).
4	Mid	Within arm's length / under 1 m / seated on the pre-designated spot / on caregiver's lap.
3	Low	Under 2 m from robot.
2	Indifferent	More than ~2 arm's lengths / metres from

1	Absent / evasive	robot. More than ~3 arm's lengths / metres, leaves the interaction, back turned, behind the robot, or at the back of the room.
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For each block, code the closest location the child reaches; in rooms shorter than 3 m, scale the distances to the room rather than using absolute metres.

Tactile engagement (physical contact with the robot or task apparatus)

Level	Meaning	Description
5	High	Touches the robot or blocks when relevant for most of the block (~7-10 s).
4	Mid	Touches the robot or task when relevant for about half the block (~4-6 s).
3	Low	Any touch with the robot or task (≤ 3 s).
1	Absent / evasive	Does not touch, refuses to touch, leaves the room, or turns away.

Left blank for the duration where a child was instructed not to touch the robot (see the 99 rule).

Communicative engagement (verbal or non-verbal communication to or about the robot/task)

Level	Meaning	Description
5	High	Responds promptly/eagerly; frequent vocalisations, imitation, stimming, gestures (~7-10 s).
4	Mid	Responds; some vocalisations, imitation, stimming, gestures (~4-6 s).
3	Low	Hesitant or occasional response with noticeable delay; infrequent vocalisations/gestures (≤ 3 s).
1	Absent / evasive	Does not respond or refuses

despite prompting; no vocalisations or gestures; turns away or leaves.

Communication is expected to vary with the child’s age and diagnosis.

Visual engagement (direction of eye gaze, not eye contact)

Level

- 5
- 4
- 3
- 1

Visual engagement concerns whether the child looks toward the robot or to a location specified by the robot or experimenter, not whether the child makes eye contact.

Emotional engagement (affective valence)

Emotional engagement is coded on a bipolar valence scale. Score each block by the highest-magnitude affect present; if both strongly positive and strongly negative affect occur within the same block, score 0. Use multiple cues (facial expression, tone of voice, speech content, posture, gestures, movement) because the face is not always visible. Use 99 only when the child is out of frame. Stimming may be positively or negatively valenced; use other cues to decide.

Level	Meaning	Description
+2	Very positive	Laughter, squealing, open-mouthed smiling, positive stimming, clapping.
+1	Positive	Closed-mouth smiling (mouth turned upward).
0	Neutral	Disinterest; looks away or up/down instead of responding.
-1	Negative	Frowning, frequent looks away, hides face, down-turned mouth, anxious stimming.
-2	Very negative	Crying, screaming, hiding, aggression/anger.

Reliability

A second coder independently rated a subset of sessions; the target was roughly 80% agreement between coders. The double-coded proportion and the reliability statistic are reported in the manuscript and supplementary materials.

Aggregation and cleaning

Block-level codes are aggregated to produce engagement scores, with 99 treated as missing. A segment-level cleaning rule excludes a segment where more than 50% of its intervals are contiguously missing (the 50%-contiguous-missingness rule applied in the analysis pipeline). See the analysis notebook for the exact implementation.

Notes for re-use

- This summary documents the coding scheme only; no participant data are included.
- The scheme was adapted from the authors' prior engagement paradigm; cite that source when re-using.